

VANDENBERG AFB, CA, USA

WMO: 723930

Lat: 34.717N

Lon: 120.567W

Elev: 113

StdP: 99.98

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 93214

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.1	2.9	-6.3	2.2	11.0	-4.1	2.7	10.6	12.4	13.3	11.5	13.2	3.1	110	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.0	27.0	15.7	24.3	15.1	22.5	15.0	18.3	22.7	17.4	21.2	16.8	20.3	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.2	19.1	16.1	11.6	18.4	15.3	11.0	17.8	52.1	22.5	49.4	21.3	47.3	20.3	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	10.5	9.0	-1.0	33.1	2.0	3.2	-2.4	35.4	-3.6	37.3	-4.7	39.1	-6.2	41.4
			-2.5	20.3	1.8	1.8	-3.8	21.6	-4.9	22.6	-5.9	23.7	-7.3	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	11.2	11.4	11.9	12.3	13.1	13.9	15.4	15.9	16.2	15.5	13.5	11.1
	DBStd	2.98	2.79	2.88	2.30	2.37	1.94	1.72	1.54	1.56	2.27	2.80	2.85	2.62
	HDD10.0	65	18	15	7	4	0	0	0	0	0	0	3	18
	HDD18.3	1814	221	195	201	184	164	133	92	79	73	99	149	224
	CDD10.0	1325	57	54	64	72	96	118	168	182	185	169	107	52
	CDD18.3	31	0	1	1	1	1	2	2	2	9	10	3	0
	CDH23.3	337	6	14	11	21	24	21	6	14	68	111	39	3
	CDH26.7	96	0	2	2	6	9	11	1	3	20	34	8	0
(14) Wind	WSAvg	3.9	4.0	4.2	4.3	4.8	4.7	4.1	3.3	3.2	3.2	3.5	3.7	4.0
(15) Precipitation	PrecAvg	422	91	93	68	29	11	2	1	0	2	20	33	71
	PrecMax	828	415	398	320	109	62	17	7	2	11	101	114	236
	PrecMin	158	5	3	1	3	1	0	0	0	0	1	4	6
	PrecStd	167	90	84	72	28	14	5	2	1	3	24	28	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	25.6	24.7	26.1	26.3	25.2	23.5	25.0	29.5	30.7	27.5	22.9
		MCWB	12.4	14.3	14.4	15.0	14.7	16.5	17.6	18.1	17.7	16.1	14.8	11.9
	2%	DB	21.1	21.7	20.9	21.3	21.0	20.7	21.5	22.1	25.1	26.3	23.6	19.9
		MCWB	11.8	12.6	12.7	13.1	14.9	15.4	16.9	17.0	17.2	15.3	13.6	11.9
	5%	DB	18.3	18.4	18.2	18.2	18.3	18.8	20.1	20.7	22.5	23.0	20.8	17.6
		MCWB	11.7	12.3	12.5	13.1	14.0	14.8	16.1	16.5	16.7	14.7	13.2	11.7
	10%	DB	16.1	16.3	16.4	16.6	17.0	17.5	18.8	19.3	20.8	20.8	18.6	16.1
		MCWB	11.7	11.8	12.3	12.5	13.3	14.1	15.5	15.8	16.3	14.9	13.0	11.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.8	15.7	16.0	16.8	17.4	18.6	18.8	20.1	19.3	16.7	15.9
		MCDB	17.5	19.6	20.3	21.5	21.8	24.0	22.1	23.4	25.5	24.2	21.7	17.9
	2%	WB	14.2	14.6	14.6	14.7	15.6	15.9	17.3	17.4	18.5	17.7	15.6	14.9
		MCDB	16.2	18.0	18.3	19.0	19.5	19.6	20.8	20.9	22.7	21.2	19.7	16.6
	5%	WB	13.5	13.6	13.7	13.7	14.5	15.1	16.4	16.7	17.5	16.6	14.8	13.8
		MCDB	15.7	16.9	16.9	17.5	17.9	18.2	19.3	19.9	21.1	20.3	18.4	15.4
	10%	WB	12.6	12.7	12.8	12.9	13.5	14.4	15.7	16.1	16.6	15.7	14.0	12.7
		MCDB	14.9	15.8	15.6	16.0	16.5	17.0	18.3	18.8	20.0	19.3	17.3	14.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	10.0	9.6	9.2	8.1	7.4	7.0	7.4	9.0	10.7	10.9	10.3
		MCDBR	13.9	14.2	13.4	12.9	11.5	10.3	9.0	9.3	12.5	15.2	14.6	12.9
	5% WB	MCWBR	7.6	7.6	7.3	7.1	6.6	6.0	4.8	4.9	6.4	7.8	7.8	7.8
		MCWBR	9.5	10.6	10.5	11.0	10.5	9.0	8.5	8.8	10.7	11.6	11.3	8.3
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.323	0.339	0.349	0.349	0.357	0.342	0.343	0.335	0.304	0.306
		taud	2.603	2.571	2.552	2.493	2.486	2.479	2.513	2.528	2.528	2.553	2.611	2.603
	Ebn at Noon		915	936	953	947	934	929	919	932	919	901	905	888
		Edh at Noon	76	87	96	107	109	110	106	102	97	87	74	72
(44) All-Sky Solar Radiation	RadAvg		3.01	3.87	5.26	6.55	7.23	7.42	6.98	6.47	5.64	4.38	3.36	2.68
	RadStd		0.23	0.39	0.34	0.40	0.48	0.35	0.27	0.30	0.31	0.33	0.19	0.24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.65	N/A	-1.36	+0.80	N/A	N/A	N/A	-167	+240	+76

Nomenclature: See separate page